Strings

Strings

- Strings are used to store and manipulate text.
- Strings are the sequence of character enclosed within the single quotes(‘), double quotes(“), or backticks(`)
- Strings can be created using the following the syntax

```
let name1 = “John”;
let name2 =‘Abin’;
let name3 =`Vrindha`
```

Template literals

- Template literals use backticks instead of quotes to define a string

  let name=`John`

- With template literals ,It is possible to use both single as well as double quotes inside a string

  let sentence=`The name “is” Hari’s `

- We can insert variables directly in template literals this is called String interpolation

  let a= `This is ${name}` //prints This is Hari

Template literals

```
// // template literals
let num = 10
/* template literals can be used to embed
javacript in string*/
console.log(`value of num = ${num}`)// output value of num = 10

// creating multiline string
const stringnew = `this is
// a multiline
// string`
console.log(stringnew);
/*output:
this is
a multiline
string
*/
```

Escape Sequence Characters

```
let text = “This is my “new" car.";
```

- JavaScript will misunderstand this string:
- The solution to avoid this problem, is to use the backslash escape character.
- The backslash (\) escape character turns special characters into string characters:

```
let text = " This is my \" new \" car.";
```

```
let text = "The character \\ is called backslash.";
```

Six other escape sequences are valid in JavaScript:

Code	Result
\b	Backspace
\f	Form Feed
\n	New Line
\r	Carriage Return
\t	Horizontal Tabulator
\v	Vertical Tabulator

String concatenation

- We can concatenate two or more string using ‘+’ symbol
- We Already discussed in the previous class.

Type casting

- Number will be typecast into string we can also use toString() to change Number into String Number() can be used to change string to Number



```
// typecasting

const num = 10
const str = "9"
console.log(str + num)//910

const num2 = 40

const numToStr = toString(num2)
console.log(typeof numToStr)//string
// typeof returns the data type of the variable

const str3 = "10"
const num3 = 10
console.log(num2+Number(str3))//output :20
```

String Properties and Methods

```
let sname = "Manu"


console.log(sname.length);  //Prints the length
of the string
console.log(sname.toUpperCase());//prints MANU
console.log(sname.toLowerCase());//prints manu
```

Access String Characters

- You can access the characters in a string in two ways.

1. One way is to treat strings as an array. For example,

```
const a = 'hello’;
console.log(a[1]); // "e“
```

2. Another way is to use the method charAt(). For example,

```
const a = 'hello’;
console.log(a.charAt(1)); // "e"
```

```
// String method

const str = "this is a example text"

console.log(str.length)//output: 22
// .length gives the length of string

// accessing the character in string:

//using [] this takes index (zero based index)
console.log(str[8])//output : a

//using charAt(index)
console.log(str.charAt(8))//output : a
```

String Properties and Methods

```
let name1 ="Ann M"
let newname = name1.replace("M","Luka")
console.log(name1); //Ann M
console.log(newname);//Ann Luka

let s_name ="Sagar"
let m_name="Alias"
let l_name="Jacky"
let full_name=s_name.concat(m_name+l_name)
console.log(full_name);//SagarAliasJacky
```

Tasks

1 What will be output the following print statement in js

console.log("hari\"".length)

2 Explore the includes, startswith and endswith function of a string

3 Extract the amount out of the string

"Please give Rs 10000"

4 Try to Change 4th character of a given string

"Where you able to do it?"

5 Note on All string methods with programs

Continue to next slide…

Tasks

6 Write a function to check if two strings are anagrams of each other. Anagrams have the same characters but in a different order. For example, "listen" and "silent" are anagrams.

7 Write a function to count the number of vowels and consonants in a given string.

8 Write a function that capitalizes the first letter of each word in a sentence. For example, "hello world" should become "Hello World.“

9 Implement a function to perform basic string compression using the counts of repeated characters. For example, the string "aabccccccaaa" would become "a2b1c5a3.“

10 Implement a basic string compression without creating a separate function. Use loops and conditional statements to create the compressed string.